

SEMESTER S5

COMPUTER NETWORKS

(Common to CS/CD/CM/CR/CA/AD/AI/CB/CN/CU/CI)

Course Code	PCCST501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce the core concepts of computer networking.
2. To develop a big picture of the internetworking implementation on Linux-based systems.
3. To impart an overview of network management concepts.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of the Internet, Protocol layering (Book 1 Ch 1) Application Layer: Application-Layer Paradigms, Client-server applications - World Wide Web and HTTP, FTP. Electronic Mail, DNS. Peer-to-peer paradigm - P2P Networks, Case study: BitTorrent (Book 1 Ch 2)	6
2	Transport Layer: Services, Protocols, UDP, TCP (Book 1 Ch 3). <i>Hands-on: Sockets Introduction, Elementary TCP Sockets, TCP Client/Server Example, I/O Multiplexing: The select and poll Functions (Book 2 Ch 3 to 6), Elementary UDP Sockets (Book 2 Ch 8), Advanced I/O Functions (Book 2 Ch 14)</i> Network Layer: Introduction, Network-layer protocols, Unicast routing, Multicast routing - Multicasting Basics, Intra domain and inter-domain routing, Next generation IP (Book 1 Ch 4), Quality of Service (Book 1 Ch 8) <i>Hands-on: Linux Kernel Implementation of Routing Table and Caches, Routing Cache Implementation Overview, Adding new entry in the Routing Table using ip command (Book 3 Ch 14)</i>	18

3	Data-Link Layer: Data link control (DLC), Multiple access protocols (MAC), Link-layer addressing, Ethernet protocol, Connecting devices (Book 1 Ch 5) Wireless LANs, Mobile IP (Book 1 Ch 6) <i>Hands-on: Datalink Provider Interface, SOCK_PACKET and PF_PACKET (Book 2 Ch 29)</i>	11
4	SNMP, ASN.1 (Book 1 Ch 9) Physical Layer: Data and signals, Digital transmission, Analog transmission, Bandwidth utilization, Transmission media (Book 1 Ch 7)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the internetworking design in terms of protocol stack and the role of various application layer protocols	K2
CO2	Illustrate the functions of the transport layer from connectionless and connection-oriented perspectives	K3
CO3	Identify how the network layer achieves host-to-host connectivity and caters to the diverse service requirements of the host applications	K3
CO4	Explain the nuances of the data link layer design and demonstrate the various data link link layer protocols	K3
CO5	Describe the fundamental characteristics of the physical layer and understand how the physical layer supports the functionalities of the top layers	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	2										3
CO3	3	2			2							3
CO4	3	2										3
CO5	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networks: A Top-Down Approach	Behrouz A Forouzan	McGraw Hill	SIE, 2017
2	Unix Network Programming, Volume 1: The Sockets Networking API	W. Richard Stevens, Andrew M. Rudoff, Bill Fenner	Pearson Education	3/e, 2004
3	TCP/IP Architecture, design, and implementation in Linux	Sameer Seth M. Ajaykumar Venkatesulu	Wiley	1/e, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networking: A Top-Down Approach Featuring Internet	J. F. Kurose and K. W. Ross	Pearson Education	8/e, 2022
2	Computer Networks, A Systems Approach	L. L. Peterson and B. S. Davie	Morgan Kaufmann	5/e, 2011

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://nptel.ac.in/courses/106/105/106105183/

SEMESTER 5

RECOMMENDATION SYSTEM

Course Code	PCAIT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs 30 Mins
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To establish the mathematical foundations and core algorithmic paradigms of personalization technology
2. To enable the critical evaluation, securing, and modern application of recommendation systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Linear Algebra notation: Matrix addition, Multiplication, transposition, and inverses; covariance matrices. Introduction: Recommender Systems Function, Applications of recommendation systems, Issues with recommender system Content-based recommendation: A High-Level Architecture of Content-based Systems, Advantages and Drawbacks of Content-based Filtering, Item Representation, Methods for Learning User Profiles.	9
2	Collaborative Recommendation: User-based nearest neighbour recommendation, Item-based nearest neighbour recommendation, model-based and pre-processing-based approaches. Attacks on collaborative recommender systems: Attack dimensions, Attack types, Countermeasures, Privacy aspects – distributed collaborative filtering	9
3	Knowledge-based recommendation: Introduction, Knowledge representation and reasoning, interacting with constraint-based recommenders, Interacting with case-based recommenders. Hybrid recommendation approaches: Opportunities for hybridization, Monolithic hybridization design, Parallelized hybridization design, Pipelined hybridization design	9

4	Evaluating recommender systems: Introduction, General properties of evaluation research, Popular evaluation designs, Evaluation on historical datasets, Alternate evaluation designs. Community-Based Web Search- The Collaborative Web Search System. Shared Web Search - The HeyStaks System, The HeyStaks Recommendation Engine. Social Tagging Recommenders Systems- Folksonomy, The Traditional Recommender Systems Paradigm, BibSonomy as Study Case, Tag Acquisition Trust and Recommendations- Introduction, Computational Trust, Trust-Enhanced Recommender Systems	12
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome	Bloom's Knowledge Level (KL)
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CO1	Explain the fundamentals of recommender systems and their applications.	K2
CO2	Apply content-based and collaborative filtering techniques.	K3
CO3	Explore knowledge-based and hybrid recommendation approaches.	K2
CO4	Evaluate recommender systems using different performance metrics.	K5
CO5	Analyze social, trust-based, and attack-resilient recommendation models.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	-
CO2	3	3	2	-	-	-	-	-	-	-	-	-
CO3	2	3	-	-	-	-	-	-	-	-	-	-
CO4	3	3	3	2	-	-	-	-	-	-	-	-
CO5	3	3	2	3	-	-	-	-	-	-	-	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Recommender Systems: An Introduction	Jannach D., Zanker M., FelFering A	Cambridge University Press	1/e, 2011
2	Recommender Systems Handbook	Ricci F., Rokach L., Shapira D., Kantor B.P drich	Springer	1/e, 2011
3	Advanced Engineering Mathematics	Erwin Kreyzig	Wiley	10/e,2011
4	Recommender Systems: The Textbook	Aggarwal C. C.	Springer	1e/2016

Video Links (NPTEL, SWAYAM...)

Module No.	Link ID
1	https://youtu.be/_hf_y-sj5Y?si=aWhWmytRzDibq88p
2	https://onlinecourses.nptel.ac.in/noc24_ge35/preview

SEMESTER 5

Web and Mobile Application for AI

Course Code	PCAIT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Mins
Prerequisites (if any)	NA	Course Type	Theory

Course Objectives:

1. To equip students with the knowledge and skills required to create, style, and script web pages using HTML5, CSS, JavaScript, and related technologies.
2. To build a strong foundation in responsive interface development across web and mobile ecosystems
3. To prepare students for professional mobile software engineering and deployment practices

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Internet Principles and Components- HTML-Style Sheets list, table, images, frames, forms-CSS-Introduction to Cascading Style Sheets-Rule-Features- Selectors- Attributes. Client-Side Programming: The JavaScript Language- JavaScript in Perspective-Syntax-Variables and DataTypes-Statements-Operators.	9
2	Fundamentals of Mobile Application Development: Introduction to Mobile Application Development, Overview of Mobile Platforms: iOS and Android, Introduction to Flutter: History, Features, and Benefits, Setting Up the Flutter Development Environment*, Mobile App Architectures (MVC, MVVM, and BLoC), Basics of Dart Programming Language.	9
3	User Interface Design and User Experience: Principles of Mobile UI/UX Design, Designing Responsive UIs with Flutter, Using Flutter Widgets: StatelessWidget and StatefulWidget, Layouts in Flutter: Container, Column, Row, Stack, Navigation and Routing in Flutter, Customizing UI with Themes and Styles.	9
4	Industry Practices and App Deployment: Advanced UI Components and Animations, App Security Best Practices, Testing and Debugging Flutter Applications, Continuous Integration/Continuous Deployment (CI/CD) with Flutter, Publishing Apps to Google Play Store and Apple App Store, Industry Trends and Future of Mobile Development with Flutter	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe structured web pages with HTML and style them using CSS techniques, including positioning, media queries	K2
CO2	Apply the basics of mobile application development and different mobile platforms, and the environment setup.	K3
CO3	Apply principles of effective mobile UI/UX design, develop responsive user interfaces using Flutter widgets.	K3
CO4	Experiment effectively with state in Flutter applications. Networking and data persistence in Flutter apps.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	-	-	-	-	-	-	3
CO2	3	3	3	3	3	-	-	-	-	-	-	3
CO3	3	3	3	3	3	-	-	-	-	-	-	3
CO4	3	3	3	3	3	-	-	-	-	-	-	3
CO5	3	3	3	3	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Web Technologies – A Computer Science Perspective”,	Jeffery Jackson	Pearson Education	2/e, 2008.
2	Flutter Cookbook	Simone Alessandria	Packt	2/e, 2023

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1.	Flutter for Beginners	Alessandro Biessek	Packt	1/e, 2019
2.	Fundamentals of Web Development	Randy Connolly, Ricardo Hoar	Pearson	1/e ,2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/nou24_cs09/preview

SEMESTER 5

REAL TIME SYSTEMS

Course Code	PBAIT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs 30 Mns
Prerequisites (if any)	PCCST403	Course Type	Theory

Course Objectives:

1. To introduce the core principles of real-time systems, scheduling architectures, and fault tolerance.
2. To equip students with the methodologies for requirements engineering and data management in real-time environments.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	REAL TIME SYSTEM AND SCHEDULING Introduction– Structure of a Real Time System –Task classes – Performance Measures for Real Time Systems – Estimating Program Run Times – Issues in Real Time Computing – Task Assignment and Scheduling – Classical uniprocessor scheduling algorithms –Fault Tolerant Scheduling.	9
2	SOFTWARE REQUIREMENTS ENGINEERING Requirements engineering process –Types of requirements – Requirements specification for real time systems – Formal methods in software specification – Structured Analysis and Design – Object oriented analysis and design and unified modelling language – Organizing the requirements document – Organizing and writing documents – Requirements validation and revision	9
3	REAL TIME DATABASES Basic Definition, Real time Vs General purpose databases, Main Memory Databases, Transaction priorities, Transaction Aborts, Concurrency control issues, Disk Scheduling Algorithms, Databases for Hard Real Time System.	8
4	CLOCK SYNCHRONIZATION AND CASE STUDIES Clock Synchronization –Clock, A Non fault– Tolerant Synchronization Algorithm – Impact of faults – Fault Tolerant Synchronization in Hardware – Fault Tolerant Synchronization in software. CASE STUDIES - Air traffic controller system.	10

Suggestion on Project Topics

1. Real-Time Operating Systems (RTOS) for Embedded Systems

- **Project Idea:** Develop and implement an embedded system using an RTOS (e.g., FreeRTOS, VxWorks). Focus on real-time task scheduling, inter-process communication, and managing time constraints for the system.
- **Key Concepts:** Task scheduling, time partitioning, inter-process communication (IPC), system initialization.
- **Applications:** Robotics, automotive systems, medical devices, IoT.

2. Design of a Real-Time Scheduling Algorithm

- **Project Idea:** Design and implement a custom real-time scheduling algorithm, such as Rate Monotonic Scheduling (RMS) or Earliest Deadline First (EDF). Test it with different system loads and provide performance comparisons.
- **Key Concepts:** Scheduling algorithms, real-time constraints, task prioritization, deadlines.
- **Applications:** Multitasking systems, real-time control systems, aerospace systems.

3. Real-Time Traffic Management System

- **Project Idea:** Create a traffic management system that dynamically adjusts traffic lights based on real-time traffic data (e.g., using sensors or cameras). Ensure the system meets real-time constraints for managing traffic flow.
- **Key Concepts:** Real-time data acquisition, event-driven programming, traffic light scheduling, sensor integration.
- **Applications:** Smart cities, intelligent transportation systems, automated traffic management.

4. Real-Time Data Acquisition and Monitoring System

- **Project Idea:** Develop a system for real-time data acquisition, such as monitoring temperature, humidity, or pressure, with data logging, visualization, and alerting features. Implement real-time constraints to ensure timely data capture and processing.
- **Key Concepts:** Real-time data collection, sensor interfacing, event-driven architecture, time-critical data processing.
- **Applications:** Industrial automation, environmental monitoring, IoT-based monitoring systems.

5. Real-Time Video Streaming System

- **Project Idea:** Design a real-time video streaming system that supports live video feeds with low latency. Implement real-time constraints to handle video encoding, transmission, and display.
- **Key Concepts:** Video compression, network protocols, latency optimization, real-time streaming.
- **Applications:** Live broadcasting, video surveillance, remote collaboration tools.

6. Real-Time Embedded System for Automotive Control

- **Project Idea:** Develop an embedded control system for an automotive application (e.g., braking, engine control). Implement a real-time operating system (RTOS) to handle critical tasks with time constraints.
- **Key Concepts:** Embedded systems, real-time control, safety-critical systems, CAN bus.
- **Applications:** Autonomous vehicles, automotive safety systems, driver assistance systems.

7. Real-Time Multi-Agent System for Cooperative Robots

- **Project Idea:** Design a real-time system where multiple robots (agents) work together to complete a task, such as exploration or assembly. Focus on communication protocols and real-time decision-making for task allocation.
- **Key Concepts:** Multi-agent systems, distributed computing, task coordination, real-time communication.
- **Applications:** Industrial automation, robotics, cooperative drone systems.

8. Real-Time Control System for Industrial Automation

- **Project Idea:** Implement a real-time control system for an industrial process (e.g., temperature control, robotic arm control, or conveyor belt systems). Ensure that the system responds within strict timing constraints.
- **Key Concepts:** Control loops, feedback systems, real-time scheduling, process monitoring.
- **Applications:** Manufacturing, process control, robotics.

9. Real-Time Communication System in Ad-Hoc Networks

- **Project Idea:** Develop a communication protocol for real-time data transmission in an ad-hoc network, where the network topology is dynamic and nodes join and leave unpredictably. Ensure that the communication is reliable and meets real-time constraints.
- **Key Concepts:** Ad-hoc networks, communication protocols, network routing, real-time data transmission.
- **Applications:** Military communications, disaster recovery systems, vehicle-to-vehicle communication.

10. Real-Time Video Surveillance System with Object Detection

- **Project Idea:** Develop a real-time video surveillance system that uses computer vision techniques for object detection (e.g., people, vehicles) and triggers real-time alerts based on specific actions or conditions.
- **Key Concepts:** Computer vision, object detection algorithms, real-time image processing, alerting systems.
- **Applications:** Security surveillance, smart cities, automated monitoring systems.

11. Real-Time Audio Processing for Speech Recognition

- **Project Idea:** Implement a real-time speech recognition system where audio input is processed and transcribed to text in real-time. Focus on reducing latency and ensuring reliable recognition under varying conditions.
- **Key Concepts:** Signal processing, real-time audio capture, speech recognition, machine learning.
- **Applications:** Voice assistants, transcription services, hands-free control systems.

12. Real-Time Health Monitoring System

- **Project Idea:** Develop a health monitoring system that captures real-time data from wearable devices (e.g., heart rate, ECG) and sends alerts to healthcare professionals if critical thresholds are breached.
- **Key Concepts:** Wearable sensors, data transmission, real-time alert systems, healthcare applications.
- **Applications:** Telemedicine, wearable health devices, elderly care.

13. Real-Time Distributed Database for Financial Transactions

- **Project Idea:** Implement a distributed database system that supports real-time transaction processing. Focus on ensuring consistency, availability, and low-latency response times.
- **Key Concepts:** Distributed systems, transaction processing, database management, consistency models.
- **Applications:** Online banking, e-commerce, real-time financial analysis.

14. Fault-Tolerant Real-Time System

- **Project Idea:** Develop a fault-tolerant real-time system that ensures high availability and correct behavior even in the presence of hardware or software failures. Implement redundancy techniques to guarantee system uptime.
- **Key Concepts:** Fault tolerance, redundancy, error handling, real-time recovery techniques.
- **Applications:** Critical systems, aerospace, military applications, automotive safety.

15. Real-Time Voice over IP (VoIP) System

- **Project Idea:** Design a real-time VoIP system that ensures high-quality voice communication with low latency, even under network congestion or packet loss.
- **Key Concepts:** Real-time voice encoding/decoding, network protocols (e.g., SIP, RTP), QoS (Quality of Service), latency optimization.
- **Applications:** Communication systems, video conferencing, remote collaboration.

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply principles of real-time system design techniques to develop real-time applications	K3
CO2	Make use of the database in real-time applications.	K2
CO3	Make use of the architectures and behaviour of real-time operating systems.	K2
CO4	Evaluate the synchronization techniques present in a real-time system.	K5
CO5	Analyse case studies and real-time project systems	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	✓	✓	✓	✓	-	-	-	-	-	-	-	✓
CO2	✓	✓	✓	✓	-	-	-	-	-	-	-	✓
CO3	✓	✓	✓	✓	-	-	-	-	-	-	-	✓
CO4	✓	✓	✓	✓	✓	✓	-	-	-	-	-	✓
CO5	✓	✓	✓	✓	✓	✓	-	-	-	-	-	✓

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Real Time Systems”	C.M. Krishna, Kang G. Shin	Tata McGraw - Hil	2010
2	Real-time systems: theory and practice	Rajib Mall	Pearson Education,	2009
3	Real-Time System Design and Analysis	Philip.A.Laplante	Prentice Hall of India	3rd Edition, 2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hard real-time computing systems: predictable scheduling algorithms and applications	Giorgio C. Buttazzo	Springer	2008
2	Resource management in real-time systems and networks	C. Siva Ram Murthy, G. Manimaran	PHI	2009
3	Real-Time Systems	C.M. Krishna, and G. Shin	McGraw-Hill International Editions	1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	http://nptel.ac.in/courses/106105036/
2	http://www.cse.unsw.edu.au
3	http://www.cse.unsw.edu.au/~cs9242/08/lectures/09- realtimex2.pdf
4	http://ppedreiras.av.it.pt/resources/str1112/apresentacoes_pesquis a/Real-time-CS.pdf

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project

Creativity in solutions and approaches

SEMESTER5

KNOWLEDGE ENGINEERING

Course Code	PEAIT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To establish the fundamental paradigms and methodologies of knowledge representation and engineering
2. To enable the design and evaluation of intelligent inference systems under both deterministic and uncertain conditions

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Knowledge Engineering Definition of knowledge engineering, Role of knowledge in AI, Knowledge representation languages (e.g., semantic networks, ontologies, production rules), Knowledge acquisition techniques (e.g., expert interviews, knowledge elicitation), Applications of Knowledge Engineering Expert systems: Development, applications, and limitations.	9
2	Natural language processing: Knowledge-based approaches Semantic Networks and Ontologies Semantic networks: Structure, properties, and applications, Ontologies: Definition, components, and benefits, Ontology development methodologies (e.g., ontology engineering, ontology-based systems)	9
3	Production rules: Syntax, semantics, and inference mechanisms Rule-based systems: Architecture, forward chaining, backward chaining Applications of rule-based systems (e.g., expert systems, decision support systems) Application in Robotics: Knowledge representation and planning	9
4	Uncertainty and Reasoning under Uncertainty Uncertainty in knowledge representation: Probability theory, fuzzy logic, possibility theory, Reasoning under uncertainty: Bayesian networks, probabilistic reasoning, fuzzy inference, Applications of uncertainty reasoning (e.g., medical diagnosis, risk assessment)	9

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered, and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain knowledge engineering, identify knowledge representation languages, and describe knowledge acquisition methods.	K2
CO2	Explain semantic networks and ontologies, and analyze their benefits.	K2
CO3	Describe various rule-based systems and evaluate their effectiveness.	K2
CO4	Apply probabilistic and fuzzy logic techniques and evaluate the effectiveness of uncertainty reasoning.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

[illegible]

CO2	2	3	-	-	-	-	-	-	-	-	2
CO3	2	2	-	-	-	-	-	-	-	-	2
CO4	2	3	3	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Artificial Intelligence: A Modern Approach	Stuart Russell and Peter Norvig	Pearson Education	4/e, 2021
2	An Introduction to Knowledge Engineering	Simon Kendal, Simon L. Kendal, Malcolm Creen	Springer London	1/e, 2007
3	Modeling with Rules Using Semantic Knowledge Engineering	Grzegorz J. Nalepa	Springer International Publishing	1/e, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Guide to Expert Systems	Wateman, Waterman Donald A.	Pearson	1/e, 1986
2	Building Expert Systems Principles, Procedures, and Applications	Elias M. Awad	West Publishing Company	1/e, 1996
3	Knowledge Engineering Principles, Methods, and Applications	Alfonso Pérez Gama	Nova Science Publishers	2/e, 2015
4	Knowledge Acquisition in Practice: A Step-by-step Guide	N. R. Milton	Springer London	1/e, 2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Artificial Intelligence: Knowledge Representation and Reasoning , IIT Madras Prof. Deepak Khemani https://nptel.ac.in/courses/106106140

2	Artificial Intelligence : Search Methods For Problem solving By Prof. Deepak Khemani IIT Madras https://onlinecourses.nptel.ac.in/noc24_cs88/preview
3	Decision-Making Under Uncertainty By Prof. N. Gautam Syracuse University https://onlinecourses.nptel.ac.in/noc24_mg69/preview

SEMESTER 5

HEALTH CARE ANALYTICS

Course Code	PEAIT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To teach the health data formats, health care policy and standards, and the significance and need of data analysis and data visualization.
2. To make the learner aware of the health data management frameworks and to help them use machine learning and deep learning algorithms in healthcare

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Healthcare Analysis - Overview - History of Healthcare Analysis Parameters on medical care systems- Health care policy- Standardized code sets–Data Formats – Machine Learning Foundations: Tree-like reasoning , Probabilistic reasoning and Bayes Theorem, Weighted sum approach.	8
2	Analytics on Machine Learning - Machine Learning Pipeline – Pre- processing –Visualization – Feature Selection – Training model parameter – Evaluation model : Sensitivity , Specificity, PPV ,NPV, FPR , Accuracy , ROC, Precision Recall Curves, Valued target variables – Python: Variables and types, Data Structures and containers, Pandas Data Frame : Operations – Scikit –Learn : Pre-processing , Feature Selection.	8
3	Healthcare Management - IOT- Smart Sensors – Migration of Healthcare Relational database to NoSQL Cloud Database – Decision Support System – Matrix block Cipher System – Semantic Framework Analysis – Histogram bin Shifting and Rc6 Encryption – Clinical Prediction Models – Visual Analytics for Healthcare.	10
4	Healthcare and Deep Learning - Introduction on Deep Learning – DFF network CNN- RNN for Sequences – Biomedical Image and Signal Analysis. Natural Language Processing and Data Mining for Clinical Data – Mobile Imaging and Analytics – Clinical Decision Support System.	10

Course Assessment Method (CIE: 40 marks,ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24 marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 subdivisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the health data formats, health care policy and standards	K2
CO2	Identify the significance and need for data analysis and data visualization.	K2
CO3	Explain the health data management frameworks	K2
CO4	Explain the use of machine learning and deep learning algorithms in healthcare	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	-	-	-	-	-	-	-	-	-	2
CO2	3	2	-	-	-	-	-	-	-	-	-	2
CO3	3	2	-	-	-	-	-	-	-	-	-	2
CO4	3	2	2	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Big Data Analytics in HealthCare	Kulkarni , Siarry, Singh, Abraham, Zhang, Zomaya , Baki,	Springer	1/e, 2020
2	Healthcare Analytics: From Data to Knowledge to Healthcare Improvement,	Hui Yang and Eva K. Lee,	Wiley	1/e, 2016
3	Healthcare Data Analytics	Chandan K. Reddy and Charu C Aggarwal	Taylor & Francis,	1/e, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Health Care Analysis Made Simple	Vikas Kumar,	Packt	1/e, 2018
2	Health Care Data Analysis and Management	Nilanjan Dey, Amira Ashour, Simon James Fong, Chintan Bhatl,	Academic Press	1/e, 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Exploring Survey Data on Health Care, IIT Roorkee https://nptel.ac.in/courses/109107190
2	Medical Image Analysis https://onlinecourses.nptel.ac.in/noc24_ee57/preview
3	Analytics in Healthcare Management and Administration https://www.coursera.org/learn/analytics-in-healthcare-management-and-administration
4	Big Data Analytics for Healthcare https://www.my-mooc.com/en/mooc/bigdataanalytics

SEMESTER 5

STORAGE TECHNOLOGIES

Course Code	PEAIT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs 30 Mns
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

- To master the core architectures, networking technologies, and virtualization paradigms of modern storage systems
- To enable the design of secure, resilient, and optimized data storage and business continuity solutions

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Storage Systems Introduction to Information Storage, Overview, Storage, Processing, Networking, Digital data and its types, Information storage, Key characteristics of data center and Evolution of computing platforms. Third Platform Technologies: Cloud computing and its essential characteristics, Cloud services and cloud deployment models, Big data analytics. Data Center Environment: Building blocks of a data center.	9
2	Intelligent Storage Systems and RAID Components of an intelligent storage system, Components, addressing, and performance of hard disk drives and solid-state drives, RAID, Types of intelligent storage systems, Scale-up and scale- out storage Architecture.	9
3	Storage Networking Technologies and Virtualization Fiber Channel SAN , Software-defined networking, FC SAN components and architecture, FC SAN topologies, link aggregation, and zoning, Virtualization in FC SAN environment. Internet Protocol SAN: iSCSI protocol, network components, and connectivity, Link aggregation, switch aggregation and VLAN, FCIP protocol, connectivity, and configuration. Fibre Channel over Ethernet SAN: Components of FCoE SAN, FCoE SAN connectivity, FCoE architecture.	9
4	Backup, Archive and Replication Securing Storage Infrastructure Introduction to Business Continuity, Backup architecture, Backup targets and methods, Cloud-based and mobile device backup, Data archive, Uses of replication and its characteristics, Compute-based, storage-based, and network-based replication. Information security goals, Storage security domains, Threats to a storage infrastructure, Security controls to protect a storage infrastructure.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course, students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the fundamentals of information storage management and various models of Cloud infrastructure services and deployment	K2
CO2	Illustrate the usage of advanced intelligent storage systems and RAID	K2
CO3	Interpret various storage networking architectures - SAN, including storage subsystems and virtualization	K2
CO4	Examine the different roles in providing disaster recovery and remote replication technologies	K3
CO5	Infer the security needs and security measures to be employed in information storage management	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	2	1	3	3	-	-	-	1	1	1	3
CO2	3	1	2	3	3	-	-	-	3	2	3	2
CO3	1	1	3	2	2	-	-	-	3	1	1	2
CO4	3	2	1	2	2	-	-	-	1	1	3	1
CO5	1	3	2	1	2	-	-	-	1	2	3	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Information Storage and Management	G. Somasundaram Alok Shrivastava	EMC Corporation, Wiley Publications	1/e, 2009
2	Introduction to Storage Area Networks	Jon Tate, Pall Beck, Hector Hugo Ibarra, Shanmuganathan Kumaravel and Libor Miklas	IBM - Redbooks	9/e, 2017
3	Storage Networks Explained	Wolfgang Mueller-Friedt, Rainer Wolafka, Nils Haustein, Ulf Troppens, Rainer Erkens	Wiley Publications	2/e, 2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Storage Technologies	Tammayi R. Nagale , Dr. Anand Khandare	Technical publications	1/e, 2024

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/108/106108058/
2	http://digimat.in/nptel/courses/video/106108058/L13.html